SCIENCE ACADEMIES'

SUMMER RESEARCH FELLOWSHIP PROGRAM-2025

Quantum Algorithms:

Detailed Study, Experimental Implementation and Fault Diagnosis of Quantum Logic Gates

A 2-Month Project Report is submitted in partial fulfillment of the requirements for the award of the Certificate of Completion of the Summer Internship 2025

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ABSTRACT

The project primarily focuses on understanding, implementing, and testing some of the most important concepts and algorithms in quantum computing. The work consists of the basics of qubits, Hilbert spaces, and quantum gates, which form the foundation of quantum computation. With these building blocks, key algorithms such as the Deutsch, Deutsch–Jozsa (DJ), and Bernstein–Vazirani(BV) algorithms were studied in detail. These algorithms were chosen because they show how quantum systems can outperform classical ones by using superposition, entanglement, and parallelism.

The project goes beyond theory by using Qiskit to simulate quantum circuits and execute them on real IBM quantum computers, namely IBM-Brisbane (Qubits 127; QPU version-1.1.202; Processor type- Eagle r3; Basis gates- ecr, id, rz, sx, x; Median readout error 2.136E-2) and IBM-Torino (Qubits 133; QPU version 1.0.90; Processor type- Heron r1; Basis gates- cz, id, rx, rz, rzz, sx, x; Median readout error 3.223E-2). More information *click here*. Several gates, including the X, NOT, CNOT, and Hadamard gates were experimentally tested. Their outputs were verified against theoretical expectations, giving confidence in both the simulator and hardware. Experiments also explored entanglement and quantum parallelism, showing how qubits can be correlated and functions can be evaluated across many inputs at once.

The algorithms were implemented using different oracle types (including both constant and balanced types). On simulators, results were ideal, while on real IBM Quantum devices, small deviations due to noise and various gate errors were observed. A stability-study compared IBM-Brisbane and IBM-Torino QCs, showing both are reliable but with slight performance differences.

A major part of the project examined faults in the Hadamard gate, focusing on coherent phase flip errors. Theoretical calculations and Bloch sphere representations were carried out, followed by experiments to detect faulty gates. Distinct patterns in output states served as clear indicators of such faults. A generalized faulty Hadamard gate with a tunable parameter θ was also studied, and fidelity calculations quantified how close faulty outputs were to the ideal case.

Overall, this work bridges theory and experiment in quantum computing. It demonstrates how quantum algorithms operate, how real devices behave under noise, and how errors in gates can be detected and analyzed. The project highlights both the power and the present limitations of quantum hardware, insights of error diagnosis and the future of reliable quantum computation.

OBJECTIVES

The purpose of this project is to explore the world of quantum computing through both theoretical study and experimental work. Quantum computers are no longer just an idea of the future—they are steadily growing into real machines that promise to solve problems classical computers cannot handle. This report brings together a careful theoretical study and experimental outcome of quantum algorithms, hands-on implementation on simulators and real QC devices, and the analysis of faults that naturally arise in quantum systems. The aim is not only to understand how these algorithms work in theory but also to see how they behave in practice, including the challenges of noise and gate imperfections.
More specifically, the objectives are:

- To study the basic building blocks of quantum computing such as qubits, qutrits, quantum logic gates, circuits and quantum registers.
- To analyze fundamental quantum algorithms like the Deutsch, Deutsch–Jozsa, and Bernstein–Vazirani algorithms, and understand how they demonstrate quantum advantage.
- To implement these algorithms using Qiskit on both simulators and IBM's real quantum computers.
- To compare theoretical predictions with experimental outcomes, highlighting the role of quantum noise and hardware limitations.
- To investigate the stability of quantum computers by repeating experiments across devices and checking consistency.
- To analyze the behavior of a faulty Hadamard gate in the Deutsch–Jozsa algorithm, both mathematically and experimentally, and propose ways to detect such faults.
- To calculate fidelity measures for assessing the closeness of experimental results with ideal states.

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Chapter 1

Introduction to Quantum Computing

Quantum computing leverages the principles of quantum mechanics to perform computations that classical computers cannot efficiently handle. Unlike classical bits that exist in states 0 or 1, quantum bits (qubits) can exist in a superposition of both states simultaneously. This property, along with entanglement, quantum parallelism, and interference, allows quantum computers to solve specific problems exponentially faster than classical computers.

1.1 Qubits, Qudits and Qutrits:

A qubit state is represented as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1.$$

It can also be expressed as:

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle,$$

where θ and ϕ define a point on the Bloch Sphere.
A qudit generalizes qubits to d levels. A qutrit has 3 basis states:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle + \gamma|2\rangle, \quad |\alpha|^2 + |\beta|^2 + |\gamma|^2 = 1.$$

1.2 Hilbert Space & Quantum Registers:

Quantum states exist in a complex vector space called Hilbert space. The Hilbert space is *linear*, which lays the foundation for the quantum superposition principle. The inner product provides for the orthogonality of the linear state space. It helps to visualize the geometry of the finite or infinite number of coordinate axes orthogonal to each other.

An n-qubit quantum register spans 2^n dimensions, allowing exponential growth in computational space. A Quantum Register is only allowed to store information about the basis states (just like the classical register), and its linear combination. A n-qubit register stores information in $\mathbb{C}^{2n}$ Hilbert Space, which is a product space of n 1-qubit systems ($\mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathbb{C}^2$......).

1.3 Quantum Gates and Cirucits:

- **X (NOT) Gate:** Flips the basis state. $X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
- **Z (Phase) Gate:** Applies a Phase difference between basis states by π. $Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

- **Walsh-Hadamard Gate:** Transforms basis state to a superposition of basis states. $H = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
- **CNOT Gate:** A 2-qubit gate for conditional flips. When Control qubit $|1\rangle$ target is flipped.
- **Toffoli/CCNOT Gate:** A 3-qubit controlled-controlled-NOT gate.
- **Quantum Circuit:** A Quantum circuit has to be read from left to right. Each line in the circuit represents a wire (not physical) in the circuit. It might corresponds to passage of time or movement of a particle such as a photon with respect to time. Quantum circuit don't allow loops. So, any quantum circuit is acyclic.

1.4 Basic Concepts of Quantum Computing:

Quantum Superposition: A qubit if exists in multiple basis states simultaneously until it is measured.

Quantum Entanglement: A Non-separable states like Bell states $\frac{|00\rangle+|11\rangle}{\sqrt{2}}$. Can't be represented by a tensor product of 2 states.

Quantum Oracle: It's a black box of unitary,reversible operations used in quantum algorithms to encode a function f such that the function(f) is not explicitly known,but the oracle let user to query it. 2 types of Quantum Oracle are there: XOR Oracle and Phase Oracle.

Quantum Parallelism: Evaluation of $f(x)$ for all x simultaneously using superposition. This property is often used in quantum algorithms to define a global property of the function.

Quantum Interference: Probability amplitudes add or cancel based on phase, enabling constructive or destructive outcomes.

Chapter 2

Quantum Algorithms for 2-level Qubit Systems

2.1 Deutsch Algorithm

PROBLEM STATEMENT

Input: A black box (Quantum Oracle) for computing an unknown function

$$f : \{0, 1\} \to \{0, 1\}$$

Promise: The function f is either Constant or Balanced.
Goal: Determine whether f is constant or balanced using a single query to the oracle.

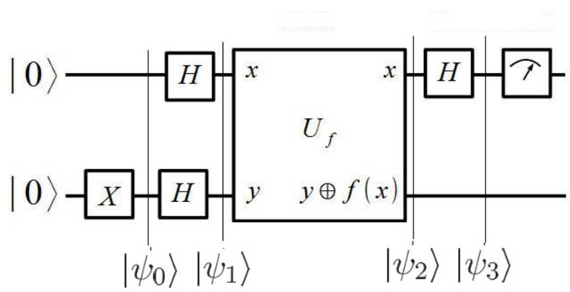


Figure 2.1: Quantum Circuit for Deutsch Algorithm

The Deutsch algorithm demonstrates quantum advantage by determining the global property of $f(x)$ with only one oracle query. The quantum circuit (Figure) begins with the initial state:

$$|\psi_0\rangle = |0\rangle |1\rangle$$

Applying Hadamard gates produces a superposition:

$$|\psi_1\rangle = \frac{1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle)$$

The oracle U_f is defined as:

$$U_f |x\rangle |y\rangle = |x\rangle |y \oplus f(x)\rangle$$

This operation introduces a phase factor $(-1)^{f(x)}$ to the first qubit (*phase kickback*). After applying $H \otimes I$, the state becomes:

$$|\psi_{\text{out}}\rangle = \frac{1}{2}\Big[|0\rangle \left((-1)^{f(0)} + (-1)^{f(1)}\right) + |1\rangle \left((-1)^{f(0)} - (-1)^{f(1)}\right)\Big] |-\rangle$$

Depending on $f(x)$:

$$\begin{aligned} f(0) = f(1) \quad &\Rightarrow \quad |\psi_{\text{out}}\rangle = \pm |0\rangle |-\rangle \quad \text{(Constant)} \\ f(0) \neq f(1) \quad &\Rightarrow \quad |\psi_{\text{out}}\rangle = \pm |1\rangle |-\rangle \quad \text{(Balanced)} \end{aligned}$$

Summary of Key Points:

- Quantum interference on the first qubit distinguishes constant and balanced functions.
- Phase kickback embeds function information into the phase of the control qubit.
- Quantum parallelism allows evaluation of both $f(0)$ and $f(1)$ simultaneously.
- Only one oracle query is needed in quantum computing versus at least two in classical computing.
- If f is *constant*, the algorithm yields state $|00\rangle$, otherwise if f is *balanced* then it yields any state other than $|00\rangle$.

2.2 Deutsch–Jozsa (DJ) Algorithm

PROBLEM STATEMENT

Input: Oracle for unknown function $f : \{0,1\}^n \to \{0,1\}$
Promise: f is either Constant or Balanced.
Goal: Identify the type of f in one query.

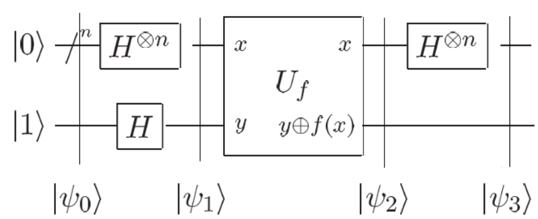


Figure 2.2: Quantum Circuit for Deutsch-Jozsa Algorithm

Algorithm:

1. Initialize state: $|\psi_0\rangle = |0\rangle^{\otimes n} |1\rangle$
2. Apply Hadamard gates:
$$|\psi_1\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right)$$
3. Oracle U_f introduces phase kickback:
$$|\psi_2\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} (-1)^{f(x)} |x\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right)$$

4. Apply Hadamard on first n qubits:

$$|\psi_3\rangle = \frac{1}{2^n} \sum_{z\in\{0,1\}^n} \left[\sum_{x\in\{0,1\}^n} (-1)^{f(x)+x\cdot z} \right] |z\rangle\, |-\rangle$$

The amplitude of the state $|0\rangle^{\otimes n}$ is:

$$A(0) = \frac{1}{2^n} \sum_{x\in\{0,1\}^n} (-1)^{f(x)}$$

Interpretation:

- If f is *constant*: All terms add constructively (± 1), so the probability of measuring $|0\rangle^{\otimes n}$ is 1.
- If f is *balanced*: Half the terms cancel out, resulting in zero probability of $|0\rangle^{\otimes n}$; a non-zero state is observed instead.
- Quantum advantage: Only 1 query is required, whereas a classical algorithm would need $2^{n-1} + 1$ queries in the worst case.

2.3 Implementation of Deutsch–Jozsa (DJ) Algorithm for 2-Qubit Input Register

PROBLEM STATEMENT

Input: Oracle for unknown function $f : \{0,1\}^2 \to \{0,1\}$
Promise: f is either Constant or Balanced.
Goal: Identify the type of f in one query.

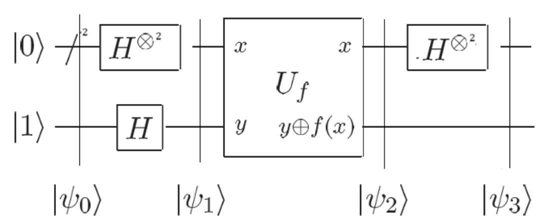


Figure 2.3: Quantum Circuit for Deutsch-Jozsa Algorithm for 2 qubit register

Algorithm:

1. **Initialize:**

$$|\psi_0\rangle = |00\rangle\, |1\rangle$$

2. **Applying Hadamard:**

$$|\psi_1\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)\, |-\rangle$$

3. **Applying Quantum Oracle U_f:** Phase kickback introduces

$$|\psi_2\rangle = \frac{1}{2}\Big[(-1)^{f(00)}\, |00\rangle + (-1)^{f(01)}\, |01\rangle + (-1)^{f(10)}\, |10\rangle + (-1)^{f(11)}\, |11\rangle\Big]\, |-\rangle$$

4. **Applying Final Hadamard:**

$$
\begin{aligned}
|\psi_3\rangle = \frac{1}{4}\Big[& |00\rangle \left((-1)^{f(00)} + (-1)^{f(01)} + (-1)^{f(10)} + (-1)^{f(11)}\right) \\
& + |01\rangle \left((-1)^{f(00)} - (-1)^{f(01)} + (-1)^{f(10)} - (-1)^{f(11)}\right) \\
& + |10\rangle \left((-1)^{f(00)} + (-1)^{f(01)} - (-1)^{f(10)} - (-1)^{f(11)}\right) \\
& + |11\rangle \left((-1)^{f(00)} - (-1)^{f(01)} - (-1)^{f(10)} + (-1)^{f(11)}\right) \Big]
\end{aligned}
$$

Result:

- If f is constant $\Rightarrow$ Constructive interference at $|00\rangle$ (probability 1).
- If f is balanced $\Rightarrow$ Constructive interference at a state other than $|00\rangle$.
- Measurement outcome determines the function type in a single query.

2.4 Bernstein-Vazirani Algorithm:

PROBLEM STATEMENT

Input: A black box (Quantum Oracle) for computing an unknown function

$$f : \{0,1\}^n \rightarrow \{0,1\}$$

Promise: The function (f) encoded in the oracle is of the form f(x)=a.x (mod 2).
Where,
a=$(a_0, a_1....a_{n-1})$$\in \{0,1\}^n$ is a hidden string.
a.x is the bitwise dot product modulo 2.
Problem: Find the hidden string 'a' with the minimum number of queries to f.

Quantum circuit for implementing the algorithm is somewhere similar to the DJ algorithm for n-qubit register.

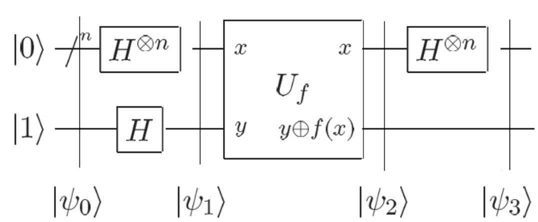


Figure 2.4: Quantum Circuit for Berstein-Vazirani Algorithm

The initial state is:

$$|\psi_0\rangle = |0\rangle^{\otimes n} |1\rangle$$

Applying $H^{\otimes n} \otimes H$ to the input state:

$$|\psi_1\rangle = H^{\otimes n} |0\rangle^{\otimes n} H |1\rangle$$

$$|\psi_1\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

The Oracle U_f is defined as:

$$U_f |x\rangle |y\rangle = |x\rangle |y \oplus f(x)\rangle$$

Now applying the Quantum Oracle U_f to the product state:

$$|\psi_2\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} \left(\frac{U_f |x\rangle |0\rangle - U_f |x\rangle |1\rangle}{\sqrt{2}} \right)$$

$$|\psi_2\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} (-1)^{f(x)} |x\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

When nth tensor of Hadamard gate is applied onto a general state $|x\rangle$ then the transformation is written as:

$$H^{\otimes n} |x\rangle = \sum_{z \in \{0,1\}^n} (-1)^{x.z} \frac{|z\rangle}{\sqrt{2^n}}$$

where, x.z is the bitwise inner product of x and z modulo 2.
Using this equation $|\psi_3\rangle$ is evaluated as:

$$|\psi_3\rangle = \frac{1}{2^n} \sum_{z \in \{0,1\}^n} \left(\sum_{x \in \{0,1\}^n} (-1)^{f(x)+x.z} \right) |z\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

As already stated f(x)=a.x then:

$$|\psi_3\rangle = \frac{1}{2^n} \sum_{z \in \{0,1\}^n} \left(\sum_{x \in \{0,1\}^n} (-1)^{a.x+x.z} \right) |z\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

$$|\psi_3\rangle = \frac{1}{2^n} \sum_{z \in \{0,1\}^n} \left(\sum_{x \in \{0,1\}^n} (-1)^{(a\oplus z).x} \right) |z\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

So,

- When a=z then the term $a \oplus z = 0$. The exponent is 0 for all x; resulting the sum= 2^n.
- When a≠z then $a \oplus z = 0$ for half of the input and 1 for rest of the half of the inputs; resulting exact half of the terms are +1 and rest half are -1. The total sum is zero.

Thus,

$$\sum_{x \in \{0,1\}^n} (-1)^{(a\oplus z).x} = 2^n \delta_{a,z}$$

Now, the final output state:

$$|\psi_3\rangle = \frac{1}{2^n} \sum_{z \in \{0,1\}^n} [2^n \delta_{a,z}] |z\rangle |-\rangle$$

$$|\psi_3\rangle = |a\rangle \otimes |-\rangle$$

(Delta function kills every term except z=a).
The pure computational basis state $|a\rangle$ when measured it yields:
1. Probability amplitude $|a\rangle$=1
2. Probability amplitude for all other $|z\rangle$=0
So, by measuring the first n qubits the algorithm always yields the hidden string a with probability 1. Thus the hidden string can be identified using a single Oracle query.

Chapter 3

Experimental Implementation and Analysis of Quantum Gates, Circuits and Algorithm

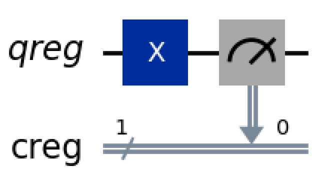
3.1 Experimental Analysis of different Quantum Gates:

3.1.1 NOT Gate

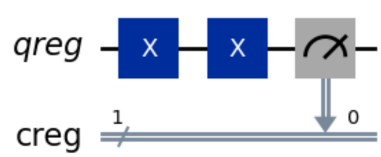
The NOT gate is implemented (Fig. 3.1) using Qiskit's QASM Simulator for two cases:

- Single NOT on $|0\rangle \rightarrow |1\rangle$
- Dual NOT on $|0\rangle \rightarrow |0\rangle$

Quantum Circuits and Simulation results (Fig. 3.2) confirm 100% probability for each outcome, validating that one NOT flips the state, while two successive NOT gates restore it, which is consistent with theory.

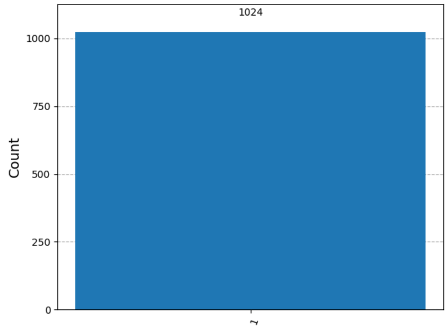


(a) Single NOT Gate implemented on state $|0\rangle$

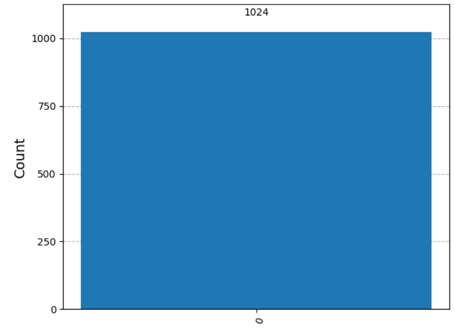


(b) Dual NOT Gate implemented on state $|0\rangle$

Figure 3.1: Application of NOT Gate and dual-NOT Gate



(a) Outcome of Circuit Fig 3.1(a) (100% $|1\rangle$)



(b) Outcome of Circuit Fig 3.1(b) (100% $|0\rangle$)

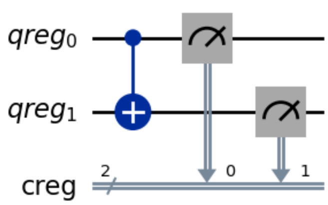
Figure 3.2: Outcome of application of NOT Gate and Dual NOT Gate

3.1.2 CNOT Gate

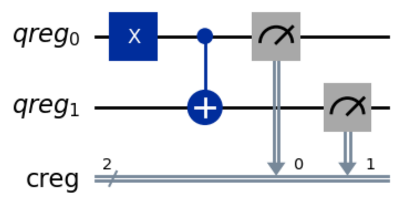
The CNOT gate was simulated using Qiskit's QASM Simulator for 1024 shots (Fig. 3.3).

- Case-1: Input $|00\rangle \rightarrow$ Output $|00\rangle$ (100%)
- Case-2: Input $|10\rangle \rightarrow$ Output $|11\rangle$ (100%)

Results shown in Fig. 3.4 confirm that the target qubit flips only when the control qubit is $|1\rangle$, matching the theoretical definition of the CNOT gate.

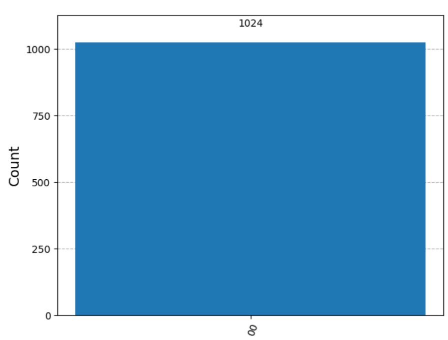


(a) Case-1

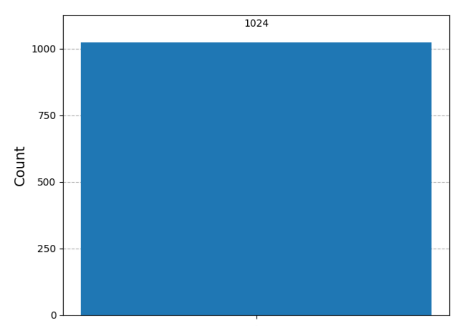


(b) Case-2

Figure 3.3: Application of CNOT gate on Quantum Registers



(a) Outcome of Circuit 3.3(a) (100% $|00\rangle$)

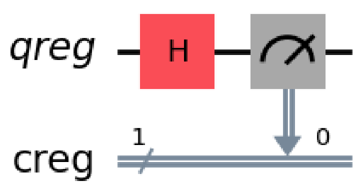


(b) Outcome of Circuit 3.3(b) (100% $|11\rangle$)

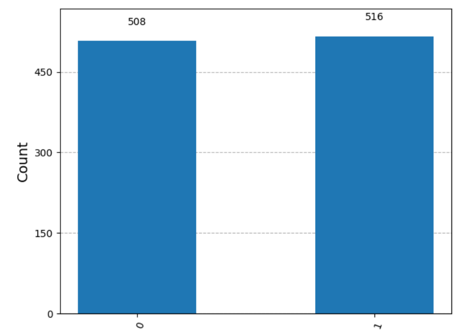
Figure 3.4: Result of application of CNOT gate on Quantum Registers

3.1.3 Walsh–Hadamard Gate

A single-qubit Hadamard transform was implemented, placing the qubit into an equal superposition. Simulation results: $|0\rangle$ observed 508 times and $|1\rangle$ 516 times ($\approx 50\%$ each), confirming the expected balanced superposition.(Fig. 3.5)



(a) Implementation of Walsh-Hadamard gate



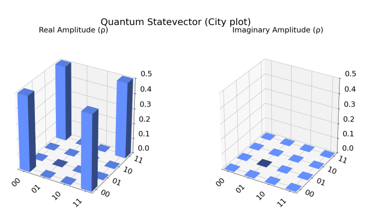
(b) Output of Walsh-hadamard transform

Figure 3.5: Implementation of Walsh-Hadamard gate on quantum register and it's simulation result

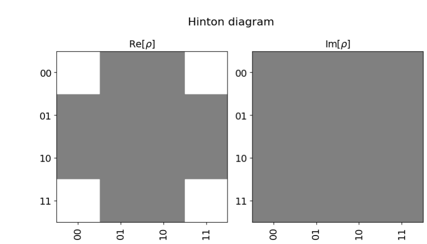
3.2 Experimental Analysis of Quantum Entanglement:

An entangling circuit (input $|00\rangle$, 4096 shots) produced $|00\rangle$ and $|11\rangle$ with nearly equal probability, confirming Bell-state entanglement.
Statevector analysis using city plot, Hinton diagram, and Bloch sphere revealed:

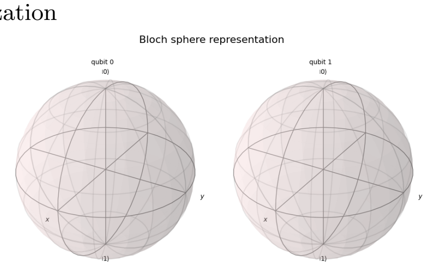
- The bars at (00,11) and (11,00) indicate Quantum Coherence in 'Cityscape' plot.(Fig. 3.6a)
- In Re(ρ) plot of Hinton Diagram no squares at diagonal entries ensures that there's coherence between the basis state i.e. the states are *entangled*.(Fig. 3.6b)
- Bloch spheres show no individual qubit polarization, implying joint entangled description.(Fig. 3.6c)



(a) Qiskit "cityscape" visualization



(b) Hinton diagram

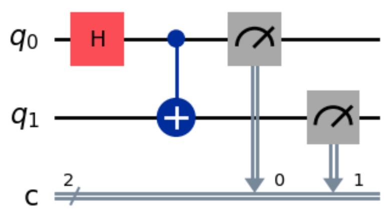


(c) Bloch sphere representation of each qubit in a Bell state

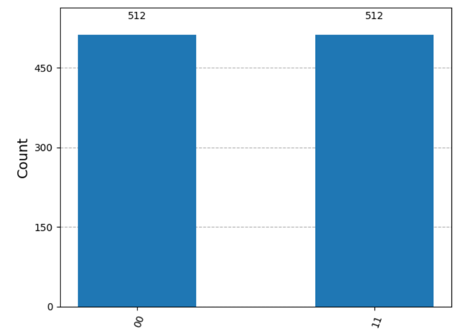
Figure 3.6: Visualization of Quantum Entanglement using various qiskit tools

3.3 Experimental Analysis of Quantum Parallelism:

Circuit of Quantum Parallelism (Fig. 3.7a), was executed on Qiskit's QASM-Simulator backend provided by the Aer module for 1024 (2^{12}) shots. The following histogram is the experimental outcome (Fig. 3.7b).



(a) Circuit for Quantum Parallelism. A Hadamard gate is used in input register 1 followed by a CNOT gate between register 1 and 2.



(b) Histogram of measurement outcomes after evaluating the function f(x)=x in superposition.The final state $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ yields outcomes $|00\rangle$ and $|11\rangle$ with equal probability

Figure 3.7: Quantum Parallelism circuit implementation and it's experimental outcome

Here, the total probability reflects the quantum state amplitudes, and they are equal.
This histogram supports the principle that, with the use of a quantum oracle in a circuit, a function can be evaluated over a superposition, of multiple inputs simultaneously. This behavior illustrates the theory of *Quantum Parallelism*.
Though circuits of Quantum Entanglement and Quantum Parallelism are similar but their purpose and interpretation are different. In case of Quantum Parallelism CNOT acts as an Oracle ($|x\rangle |y\rangle = |x\rangle |y \oplus f(x)\rangle$); while for Quantum Entanglement it acts as an entangling gate. The output of Quantum Entanglement corresponds to pairs (x, f(x)); whereas for Quantum Parallelism output shows correlation between two qubits.

3.4 Experimental Analysis of Deutsch Algorithm

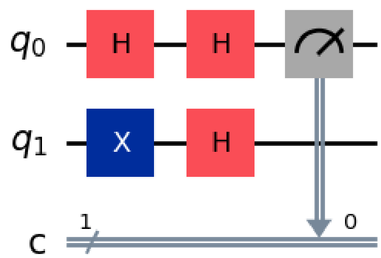
The Deutsch Algorithm was implemented using Qiskit on both the QASM Simulator and IBM's real quantum hardware (*IBM-Brisbane*) to distinguish between constant and balanced functions using a single oracle query. Four oracle types were tested:

- Type 1: No operation.
- Type 2: NOT gate on register 1.
- Type 3: CNOT (reg-0 is control qubit while reg-1 is target qubit).
- Type 4: CNOT followed by NOT on q_1.

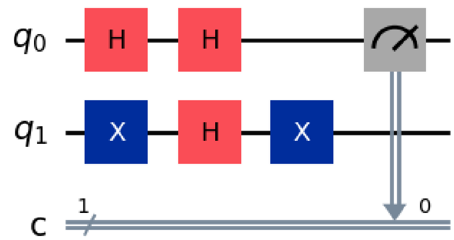
The different cirucits are shown in Fig. 3.8
On the *QASM Simulator*, all of the circuits produced ideal, noise-free results matching theoretical predictions.(Fig. 3.9) On *IBM-Brisbane*, small deviations occurred due to gate errors, decoherence, and readout noise (Fig. 3.10). All of the cases when compared with QASM Simulator results, it cues to a similar conclusion:

- Types 1 & 2 → Constant functions.
- Types 3 & 4 → Balanced functions.

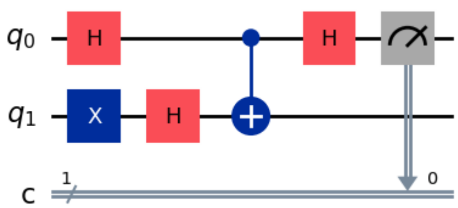
Thus, the experimental outcomes validated the Deutsch Algorithm's correctness across both simulated and real quantum hardware. The resultant outcomes are provided below:



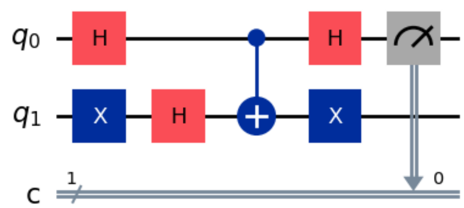
(a) Circuit for Oracle type 1



(b) Circuit for Oracle type 2

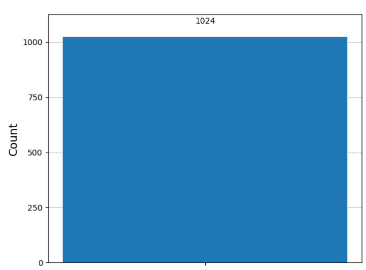


(c) Circuit for Oracle type 3

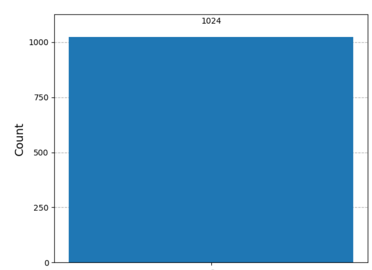


(d) Circuit for Oracle type 4

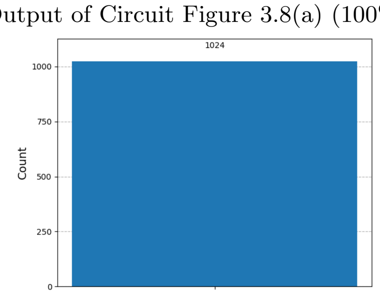
Figure 3.8: Deutsch Algorithm circuit for different Oracle types



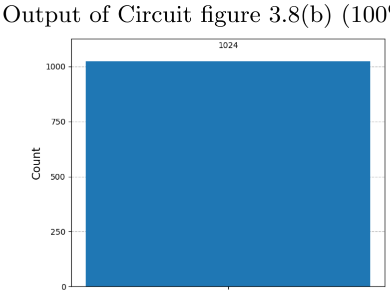
(a) Output of Circuit Figure 3.8(a) (100% $|0\rangle$)



(b) Output of Circuit figure 3.8(b) (100% $|0\rangle$)

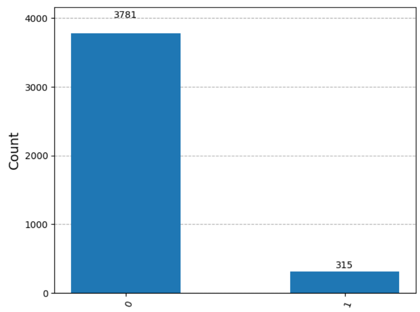


(c) Output of Circuit figure 3.8(c) (100% $|1\rangle$)

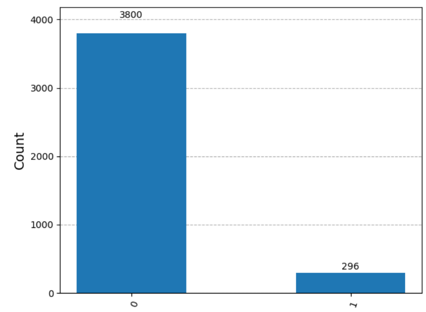


(d) Output of Circuit figure 3.8(d) (100% $|1\rangle$)

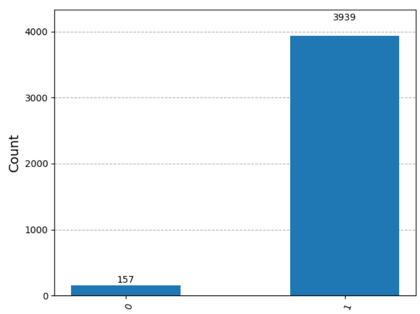
Figure 3.9: Experimental results of the Deutsch Algorithm circuits executed on the QASM Simulator for 1024 (2^{10}) shots for various function types.



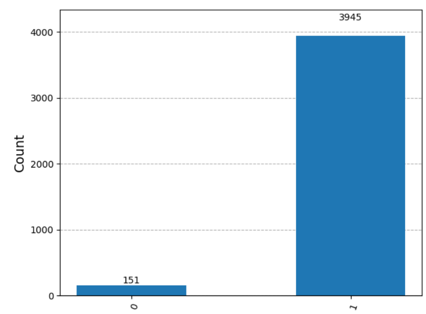
(a) Output of Circuit figure 3.8(a) (92.3% $|0\rangle$)



(b) Output of Circuit for Figure 3.8(b) (92.77% $|0\rangle$)



(c) Output of Circuit for Figure 3.8(c) (96.16% $|1\rangle$)



(d) Output of Circuit figure 3.8(d) (96.31% $|1\rangle$)

Figure 3.10: Experimental results of the Deutsch Algorithm circuits executed on the "IBM-Brisbane" quantum computer for 4096 (2^{12}) shots for various function types.

3.5 Experimental Analysis of Deutsch–Jozsa Algorithm for 2-qubit Input Register

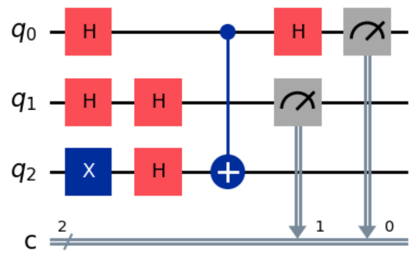
The Deutsch–Jozsa algorithm was implemented on Qiskit using both the QASM Simulator and *IBM-Brisbane* to classify a two-bit function as constant or balanced in a single query. Five different oracle types were tested:

- Types 1–3: CNOT-based oracles applied in between register 1st, nth; 2nd, nth; and both respectively.
- Type 4: No operation.
- Type 5: NOT on the target qubit.

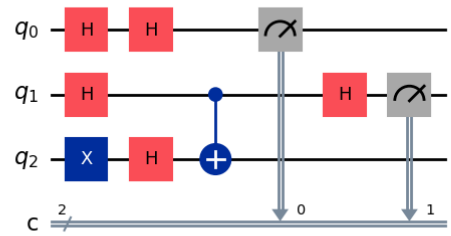
The different circuit are shown in Fig. 3.11. On the *QASM Simulator*, results were ideal and matched theoretical predictions exactly (Fig. 3.12) On *IBM-Brisbane*, small deviations arose from hardware noise and gate errors (Fig. 3.13), but the correct classification of function types was preserved:

- Types 1–3 $\rightarrow$ Balanced functions.
- Types 4–5 $\rightarrow$ Constant functions.

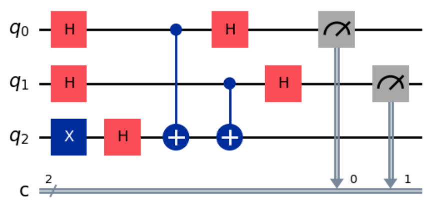
Thus, the experimental results validate the algorithm on both simulated and real quantum hardware.



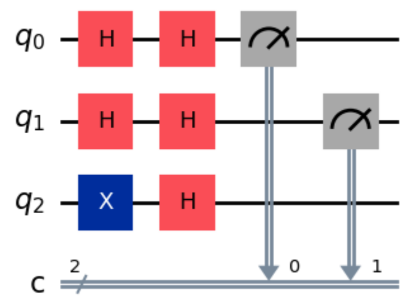
(a) Circuit for Oracle type 1



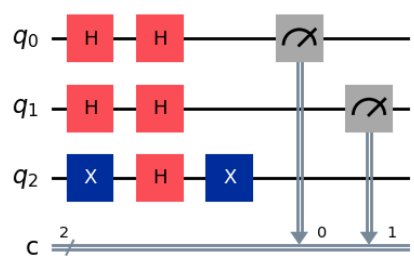
(b) Circuit for Oracle type 2



(c) Circuit for Oracle type 3

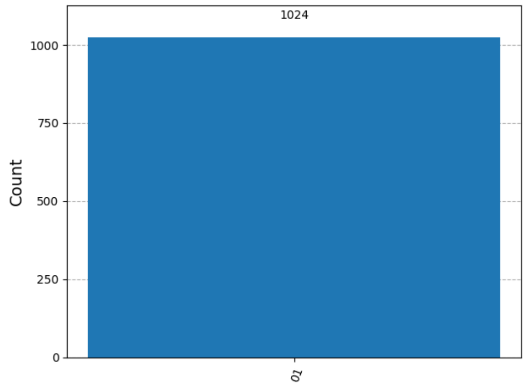


(d) Circuit for Oracle type 4

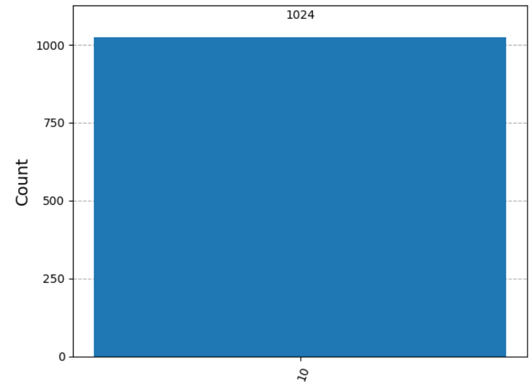


(e) Circuit for Oracle type 5

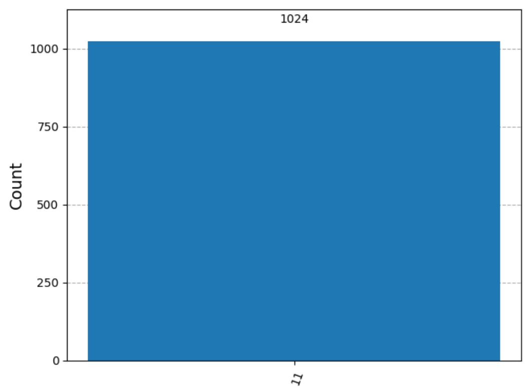
Figure 3.11: Deutsch-Jozsa Algorithm circuits implemented on 2-qubit register for different Oracle types



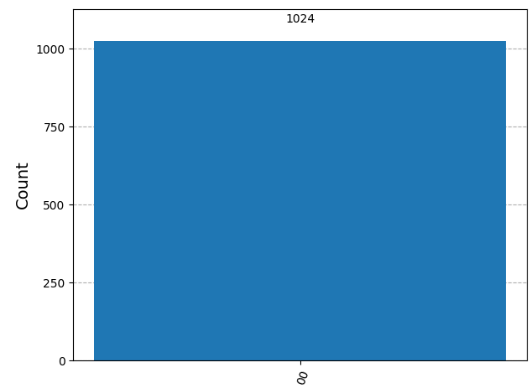
(a) Output of Circuit Figure 3.11(a) (100% $|01\rangle$)



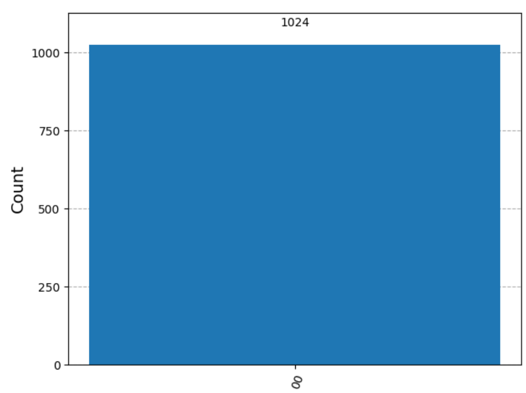
(b) Output of Circuit Figure 3.11(b) (100% $|10\rangle$)



(c) Output of Circuit Figure 3.11(c) (100% $|11\rangle$)

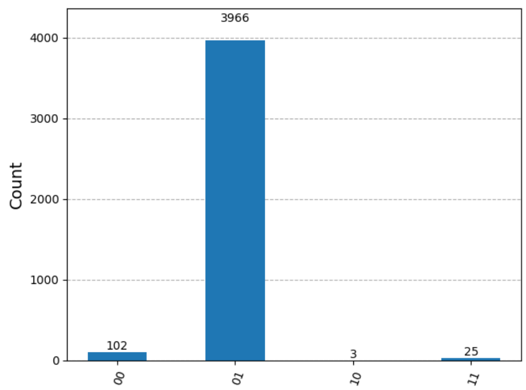


(d) Output of Circuit Figure 3.11(d) (100% $|00\rangle$)

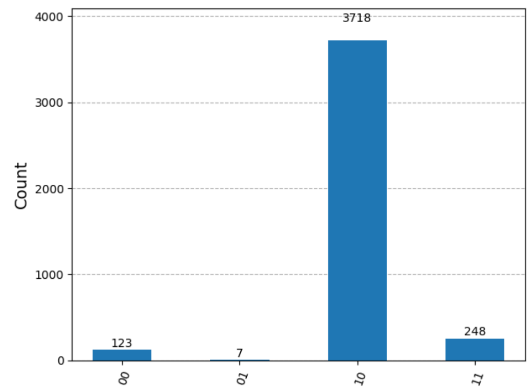


(e) Output of Circuit Figure 3.11(e) (100% $|00\rangle$)

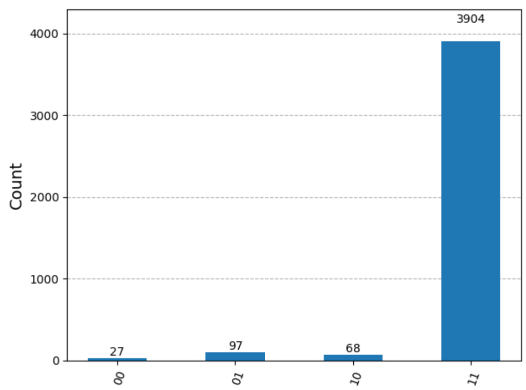
Figure 3.12: Experimental results of the Deutsch Algorithm circuits executed on the QASM Simulator for 1024 shots (2^{12}) for various function types.



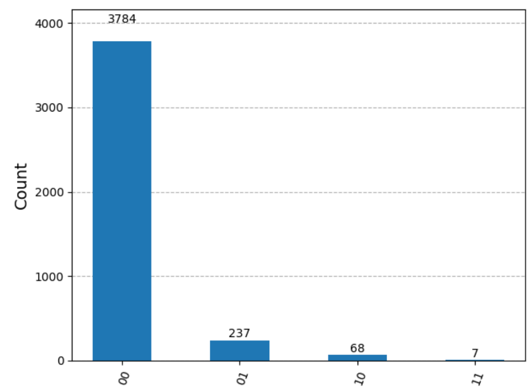
(a) Output of Circuit figure 3.11(a) (96.82% $|01\rangle$)



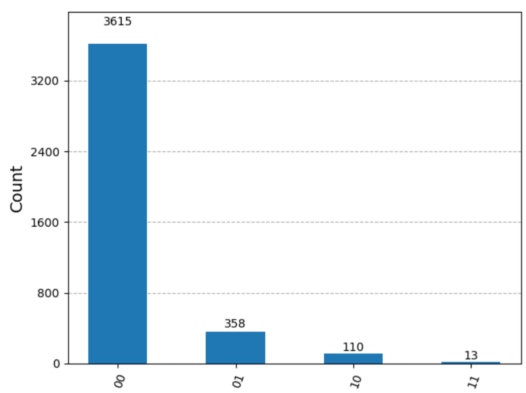
(b) Output of Circuit for Figure 3.11(b) (90.77% $|10\rangle$)



(c) Output of Circuit for Figure 3.11(c) (95.31% $|11\rangle$)



(d) Output of Circuit figure 3.11(d) (92.38% $|00\rangle$)



(e) Output of Circuit figure 3.11(e) (88.25% $|00\rangle$)

Figure 3.13: Experimental results of the Deutsch-Jozsa Algorithm circuits executed on the "IBM-Brisbane" quantum computer for 4096 shots (2^{12}) for various function types.

3.6 Stability Analysis of IBM Quantum Computers

The stability of IBM quantum computers was analyzed by implementing the Deutsch–Jozsa algorithm ($n = 2$) on two devices IBM-Brisbane and IBM-Torino.

The experimental observations are noted as following:

Number of Experiments (Unit-2^{12})	1	2	3	4	5	6	7	8	9	10
Exact (%)	100	100	100	100	100	100	100	100	100	100
Probability (IBM-Brisbane) (%)	95.7	96.6	96.2	96.7	96.5	96.2	95.6	96.2	96.3	96.3
Probability (IBM-Torino) (%)	94.9	95.5	94.7	95.7	95.2	95.9	95.5	94.8	95.0	95.5

Table 3.1: Tabulation of experimental probability of state $|11\rangle$ obtained by using Oracle-3 in DJ Algorithm (n=2) for different IBM QCs (See Fig. 3.14a).

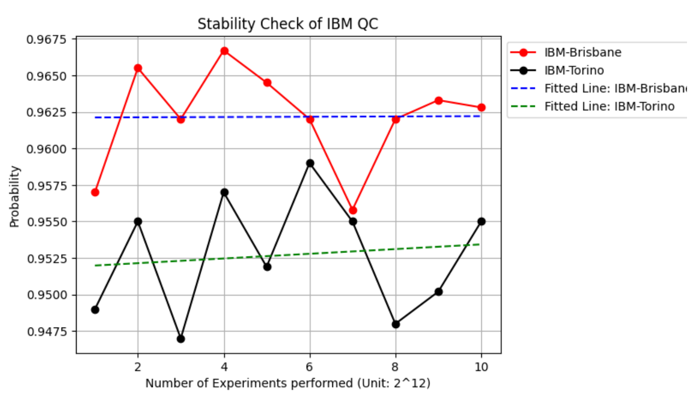
Number of Experiments (Unit-2^{12})	1	2	3	4	5	6	7	8	9	10
Exact (%)	100	100	100	100	100	100	100	100	100	100
Probability (IBM-Brisbane) (%)	88.2	90.9	90.1	90.2	90.9	91.1	92.6	94.8	87.9	88.7
Probability (IBM-Torino) (%)	87.1	87.2	86.7	86.9	87.0	87.2	86.6	86.3	87.9	86.9

Table 3.2: Tabulation of experimental probability of state $|00\rangle$ obtained by using Oracle-5 in DJ Algorithm (n=2) for different IBM QCs. (See Fig. 3.14b)

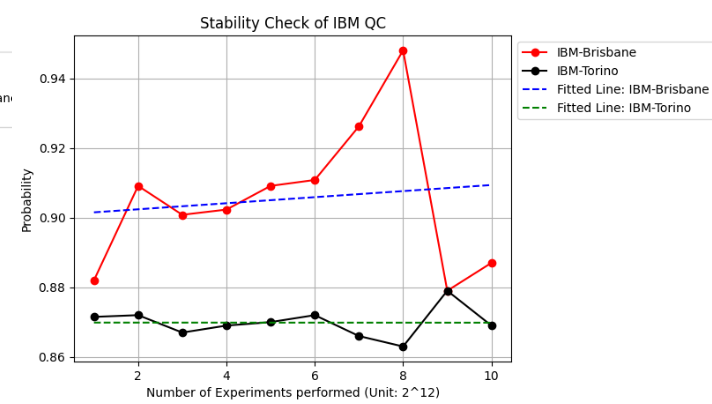
From these experimental results it's concluded that:

- Oracle-3: Average yield percentage for $|11\rangle$ state in IBM-Brisbane: 96.22%, IBM-Torino: 95.27% $\rightarrow$ Identified as *Balanced*.
- Oracle-5: Average yield percentage for $|00\rangle$ state in IBM-Brisbane: 90.55%, IBM-Torino: 86.98% $\rightarrow$ Identified as *Constant*.

Both devices demonstrated stable performance, with IBM-Brisbane slightly outperforming IBM-Torino due to lower gate and readout errors. Successful execution of the algorithm for Oracle-3 showed higher accuracy on both devices compared to Oracle-5.



(a) For Oracle-3



(b) For Oracle-5

Figure 3.14: Stability of IBM-Brisbane and IBM-Torino for different oracle types

Chapter 4

Analysis of Hadamard Gate under Coherent Phase Flip Error in DJ Algorithm for two-qubit register:

In quantum circuits, various types of errors can cause deviations from the intended output states. In this analysis, the attention is restricted to a single type of error: an imperfection in the Hadamard gates. Specifically, a *coherent phase-flip error* is considered in which the Hadamard gate transforms the computational basis states as

$$|0\rangle \longrightarrow \frac{|0\rangle - |1\rangle}{\sqrt{2}}, \quad |1\rangle \longrightarrow \frac{|0\rangle + |1\rangle}{\sqrt{2}},$$

instead of the ideal transformation.
The sole objective is to determine which Hadamard gate in the circuit is faulty by analysing the measurement statistics of the output states.

4.1 Bloch Sphere Trajectories

A single qubit, initially at state $|0\rangle$ changes it's initial state under application of Ideal Hadamard Gate and Faulty Hadamard gate. The final evolved state after application of H and H_{faulty} is shown in a comparitive Bloch sphere diagram side by side. (Fig. 4.1)

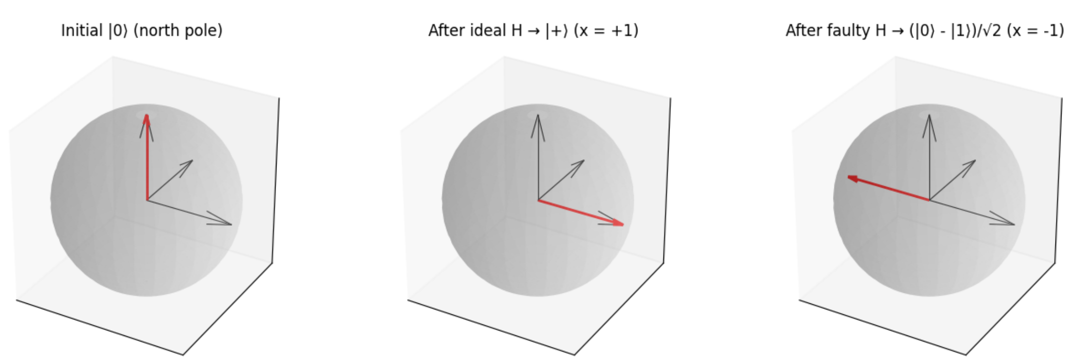


Figure 4.1: Comparative representation of state $|0\rangle$ evolution under H and H_{faulty}

4.2 Theoretical Calculation:

Firstly, it's assumed that the first Hadamard gate in the first layer of DJ Algorithm is faulty, while all other gates are ideal.
The input state:

$$|\psi_0\rangle = |00\rangle_{q_1 q_0} |1\rangle$$

Operating $H \otimes H_{faulty} \otimes H$ on the input register:

$$|\psi_1\rangle = \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right)\left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right)\left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right)$$

Operating Oracle U_f on the state $|\psi_1\rangle$:

$$|\psi_2\rangle = \frac{1}{2\sqrt{2}}\Big[|00\rangle |0 \oplus f(00)\rangle - |00\rangle |1 \oplus f(00)\rangle - |01\rangle |0 \oplus f(01)\rangle + |01\rangle |1 \oplus f(01)\rangle$$
$$+ |10\rangle |0 \oplus f(10)\rangle - |10\rangle |1 \oplus f(10)\rangle - |11\rangle |0 \oplus f(11)\rangle + |11\rangle |1 \oplus f(11)\rangle \Big]$$

Taking global phase factor into account:

$$|\psi_2\rangle = \frac{1}{2}\left[(-1)^{f(00)} |00\rangle - (-1)^{f(01)} |01\rangle + (-1)^{f(10)} |10\rangle - (-1)^{f(11)} |11\rangle\right]\left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right)$$

Imposing the final layer of Hadamard gates:

$$|\psi_3\rangle = \frac{1}{2}\Big[|00\rangle \left((-1)^{f(00)} - (-1)^{f(01)} + (-1)^{f(10)} - (-1)^{f(11)}\right)$$
$$+ |01\rangle \left((-1)^{f(00)} + (-1)^{f(01)} + (-1)^{f(10)} + (-1)^{f(11)}\right)$$
$$+ |10\rangle \left((-1)^{f(00)} - (-1)^{f(01)} - (-1)^{f(10)} + (-1)^{f(11)}\right)$$
$$+ |11\rangle \left((-1)^{f(00)} + (-1)^{f(01)} - (-1)^{f(10)} - (-1)^{f(11)}\right)\Big]\left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right)$$

From the output state $|\psi_3\rangle$ we can conclude:
1. When function f is constant:

- f(00), f(01), f(10), f(11) all of these values can be either 0 or 1.
- In both of those cases, constructive quantum interference is seen for state $|01\rangle$ and destructive quantum interference is seen for rest of the states.

2. When the function f is balanced:

- Among output f(00), f(01), f(10) and f(11), any two values are 0 and the rest two values are 1.
- In that case constructive quantum interference can be seen for any state other than $|01\rangle$. Rest of the states suffers destructive interference.

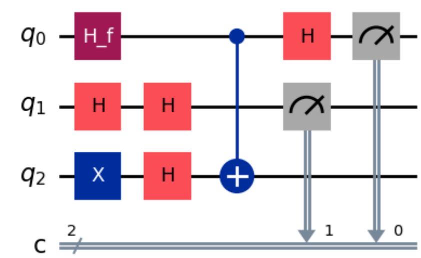
So, it's can be inferred that state $|01\rangle$ serves as an *unique fingerprint* to detect if the first hadamard of first layer is faulty.
Similarly, if the 2nd hadamard of first layer, or both of the hadmards of the first layer are faulty then respectively state $|10\rangle$ and $|11\rangle$ serves as the *unique fingerprint* to identify which hadamard gate is faulty. The detailed calculation is provided in the Appendix.

The quantum circuits and their experimental results are given respectively in the following Fig. 4.2 and Fig. 4.3 when the 1st Hadamard gate of 1st layer is faulty.

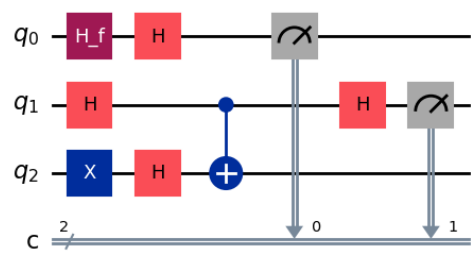
4.3 Experimental Implementation and Checking:

4.3.1 When first Hadamard of first layer is faulty:

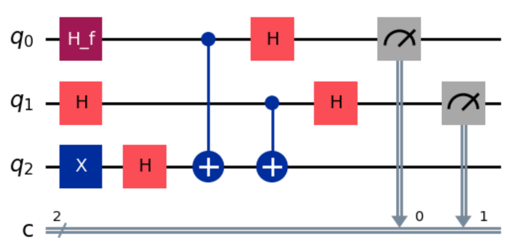
When the first Hadamard gate of the initial layer in the Deutsch–Jozsa algorithm for a 2-qubit register is faulty, the circuit is executed for the five different types of oracles considered earlier, and the corresponding experimental results are obtained as follows.



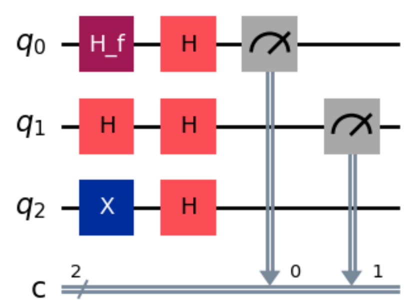
(a) Circuit for Oracle type 1



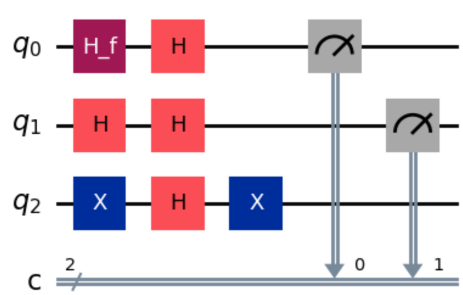
(b) Circuit for Oracle type 2



(c) Circuit for Oracle type 3

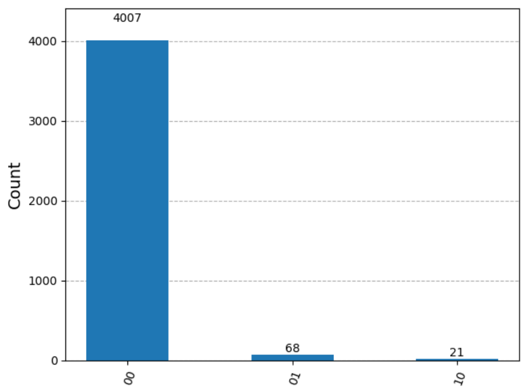


(d) Circuit for Oracle type 4

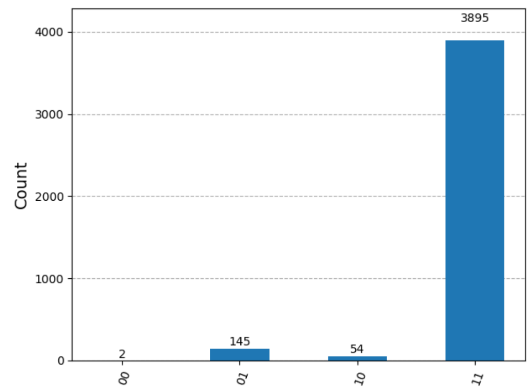


(e) Circuit for Oracle type 5

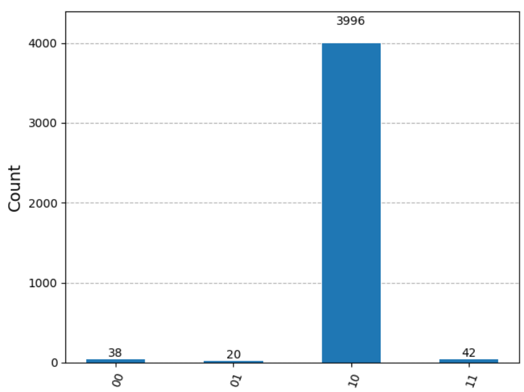
Figure 4.2: Deutsch-Jozsa Algorithm circuits implemented on 2-qubit register for different Oracle types when 1st Hadamard of input register is faulty



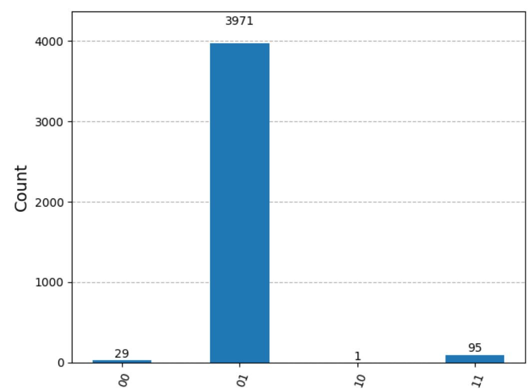
(a) Output of Circuit Figure 31(a)



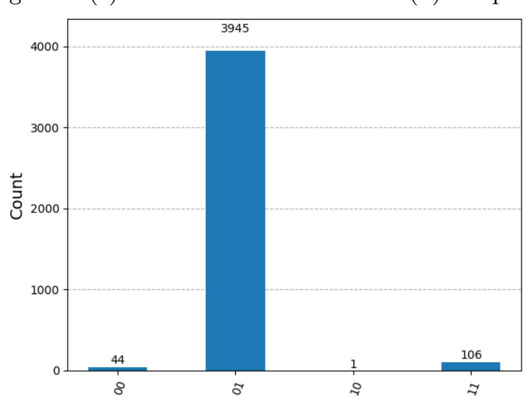
(b) Output of Circuit Figure 31(b)



(c) Output of Circuit Figure 31(c)



(d) Output of Circuit Figure 31(d)



(e) Output of Circuit Figure 31(e)

Figure 4.3: Experimental results when the Deutsch-Jozsa Algorithm circuits for 2-qubit register with Faulty hadamard at q_0 executed on the IBM-Brisbane for 4096 shots (2^{12}) for various oracle types.

The experimental outcomes can also be visualized also by the heatmap of measurement probabilities. For each case, a heatmap of output state probabilities across oracles summarizes the experimental outcomes. Using Qiskit's AerSimulator first output probabilities are generated, arranged in a matrix format to create the heatmap. In Fig. 4.4 "Yellow" shaded regions indicating the the state having maximum output probabilities; whereas "Violet" shaded regions indicating states having minimum output probabilities. Here, as ideal simulation is considered maximum probability is 1.00 and minimum probability is 0.00 stands always true.

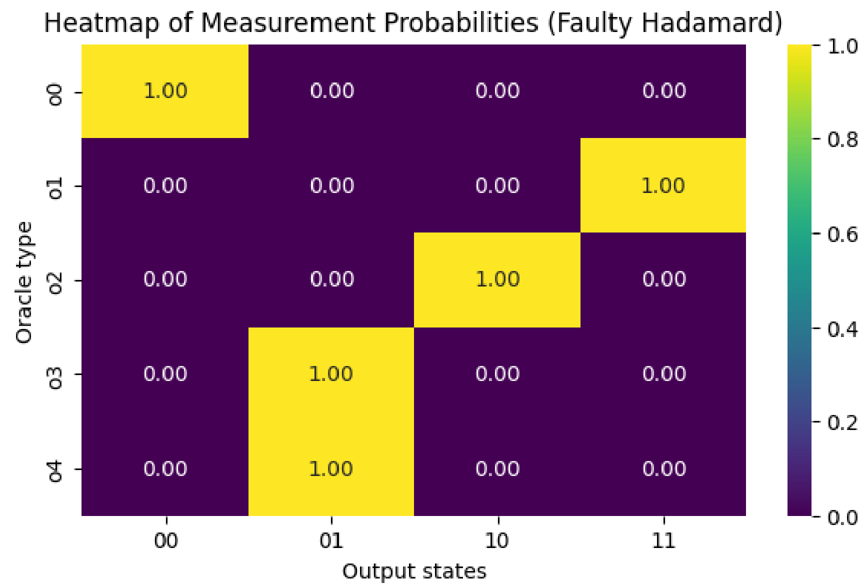


Figure 4.4: Heatmap representation of Output states in DJ Algorithm when H_{faulty} is used

In conclusion, the experimental outcomes exhibit a characteristic distribution: either the state $|01\rangle$ appears with maximum probability, or the probability of $|01\rangle$ remains minimal, maximizing any other state leaving it aside; which is in complete concordance with the theoretical calculation.
Hence, the observation of such a distribution, analogous to the appearance of the state $|00\rangle$ in the ideal case, *serves as a clear indicator* of a fault in the first Hadamard gate of the initial layer in the Deutsch-Jozsa algorithm. Measuring only the state $|01\rangle$ determines the function-type embedded in oracle.

Similarly there are other possibilities that might be the second hadamard gate of first layer is faulty or both of them are faulty. Those cases were analysed and worthy calculations along with plots are given in the Appendix section.

4.4 Generalised Faulty Hadamard Analysis

Till now, a specified version of faulty Hadamard is dealt with, whose matrix representation is:

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

If a controllable phase parameter θ is introduced in the matrix to form a generalized faulty hadamard then its matrix representation will look like:

$$H(\theta) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -e^{i\theta} & e^{i\theta} \end{bmatrix}$$

For $\theta = 0$, H(θ)= H_{ideal} and for $\theta = \pi$, H(θ)= H_{faulty}, returning the coherent phase flip case. Thus θ can interpolate between ideal and faulty hadamard cases.
It's action on computational basis states looks like:

$$H(\theta)\,|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle - e^{i\theta}\,|1\rangle)$$

$$H(\theta)\,|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{i\theta}\,|1\rangle)$$

4.5 Fidelity Calculation

In quantum information, 'Fidelity' is used to quantify how close the actual output state is to the expected ideal state. Mathematically ideal output state is the pure state represented by $|\psi_{ideal}\rangle$ whereas experimentally obtained state is $|\psi_{exp}\rangle$; then fidelity is defined as:

$$F = |\langle\psi_{ideal}|\psi_{exp}\rangle|^2$$

It calculates the squared overlap between the two states. Moreover it tells us how faithfully a quantum algorithm preserves its intended output state in presence of errors.
As stated in Sec. 2.3 the DJ algorithm for constant function ideally outputs $|\psi_{ideal}\rangle = |00\rangle$. So fidelity reduces to just the probability amplitude of $|00\rangle$ in the faulty state.

Again, it is assumed that the first Hadamard gate in the first layer of DJ Algorithm is faulty, while all other gates are ideal.
The input state:

$$|\psi_0\rangle = |00\rangle_{q_1 q_0} |1\rangle$$

Operating $H \otimes H_\theta \otimes H$ on the input register:

$$|\psi_1\rangle = \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right)\left(\frac{|0\rangle - e^{i\theta}|1\rangle}{\sqrt{2}}\right)\left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right)$$

Oracle U_f is defined as: $|x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle$
Operating Oracle U_f on the state $|\psi_1\rangle$:

$$|\psi_2\rangle = \frac{1}{2\sqrt{2}}\Bigg[|00\rangle|0\oplus f(00)\rangle - |00\rangle|1\oplus f(00)\rangle + |10\rangle|0\oplus f(10)\rangle - |10\rangle|1\oplus f(10)\rangle$$
$$+e^{i\theta}\Big(-|01\rangle|0\oplus f(01)\rangle + |01\rangle|1\oplus f(01)\rangle - |11\rangle|0\oplus f(11)\rangle + |11\rangle|1\oplus f(11)\rangle\Big)\Bigg]$$

Considering global phase factor:

$$|\psi_2\rangle = \frac{1}{2}\left[(-1)^{f(00)}|00\rangle + (-1)^{f(10)}|10\rangle + e^{i\theta}\Big(-(-1)^{f(01)}|01\rangle - (-1)^{f(11)}|11\rangle\Big)\right]\left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right)$$

As we are concerned about constant functions:

$$f(00) = f(01) = f(10) = f(11) = 0$$

So, $|\psi_2\rangle$ eventually collapses to $|\psi_1\rangle$.
Now, applying the final layer of hadamard gate:

$$|\psi_3\rangle = |0\rangle \otimes \frac{1}{2}\Big(|0\rangle + |1\rangle - e^{i\theta}(|0\rangle - |1\rangle)\Big) \otimes |-\rangle$$

$$|\psi_3\rangle = \frac{1}{2}\Big(|00\rangle(1 - e^{i\theta}) + |01\rangle(1 + e^{i\theta})\Big) \otimes |-\rangle = |\psi_{exp}\rangle \otimes |-\rangle$$

Now, the fidelity

$$F(\theta) = |\langle\psi_{ideal}|\psi_{exp}\rangle|^2 = \left(\frac{1 - e^{i\theta}}{2}\right)^2 = \frac{1}{4}(2 - 2cos\theta) = \frac{1}{2}(1 - cos\theta) = sin^2\frac{\theta}{2}$$

So,

- If $\theta = \pi$ then F(θ)=1. It means experimental state and ideal state perfectly matched. It indicates there are no such deviations from the original algorithm.
- If $\theta = 0$ then F(θ)=0. It means experimental state and ideal state are purely orthogonal to each other; there are no overlaps at all.
- If there are any kind of intermediate values of θ, F(θ) lies between 0 and 1 ($0 < F(\theta) < 1$).
- This is why, when the pre-assumed faulty Hadamard H_{faulty} is used ($H(\theta)$ when $\theta = \pi$) in the simulations, state $|00\rangle$ never appeared for constant function case. Moreover a new state other than $|00\rangle$ appeared consistently.

The evolution graph of continuous change of fidelity over the range $0 \leq \theta \leq \pi$ is given in Fig. 4.5

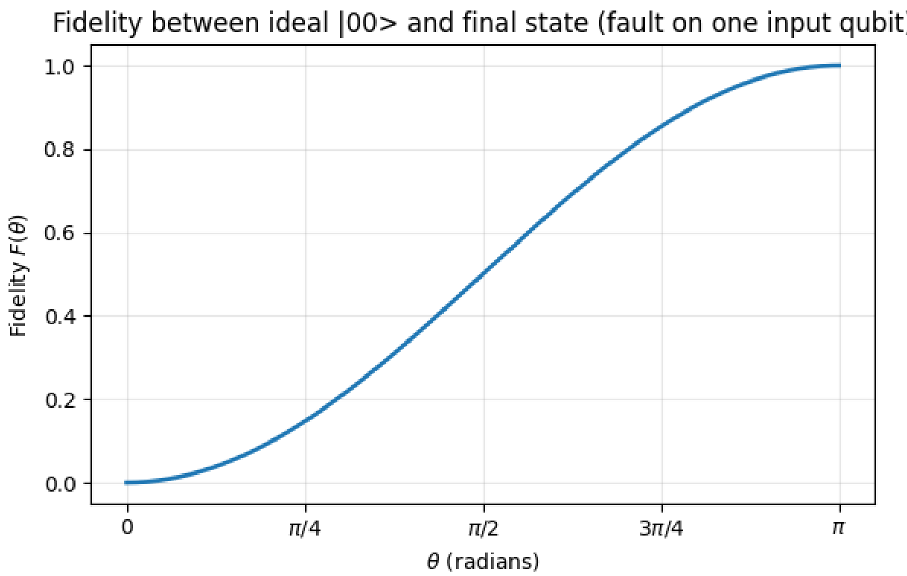


Figure 4.5: Evolution of Fidelity F(θ) over the range $0 \leq \theta \leq \pi$

Discussion & Conclusion

This study has systematically examined the fundamentals of quantum computation through both simulation and hardware implementation. Beginning with single- and two-qubit gates such as NOT, CNOT, and Hadamard, the experimental verification established the correctness of their unitary operations. The successful reproduction of entangled Bell states and demonstrations of quantum parallelism further confirmed the non-classical features of superposition, coherence, and entanglement, which are central to quantum information processing.

The Deutsch and Deutsch–Jozsa algorithms were implemented to highlight the quantum advantage of oracle-based computation. Experiments on IBM-Brisbane revealed small but systematic deviations from QASM Simulator results due to noise and decoherence. Despite these imperfections, the algorithms retained their functionality, classifying constant and balanced functions with high accuracy. A stability analysis between IBM-Brisbane and IBM-Torino confirmed consistent performance, though Brisbane exhibited slightly superior fidelity, attributable to its lower gate and readout error rates.

A key contribution of this project lies in the diagnosis of faulty Hadamard gates within the Deutsch–Jozsa framework. By introducing a coherent phase-flip error, distinct output signatures were identified, providing a diagnostic tool to localize and characterize gate imperfections. The generalization to a parameterized faulty Hadamard $H(\theta)$ allowed interpolation between ideal and erroneous behavior, offering a broader perspective on systematic error models. Fidelity analysis quantitatively captured the deviation from the ideal output state and demonstrated the sensitivity of algorithmic outcomes to small variations in phase.The integration of theoretical derivations, simulations, and experimental results provides a complete picture of algorithmic robustness under practical conditions.

In conclusion, this project not only validated fundamental quantum algorithms across different platforms but also underscored the importance of error characterization, especially in critical gates like Hadamard. Such error-aware analyses contribute to the broader effort of developing fault-tolerant quantum computation. Future work may extend these studies to multi-qubit registers, qutrit registers, explore advanced error-mitigation strategies, and test algorithmic stability under different classes of noise. Together, these efforts pave the way toward reliable quantum computation and its application to problems inaccessible to classical systems.

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This study has made extensive use of several open-source and publicly available software tools, platforms, and computational resources. In particular, the Anaconda Navigator has been employed as the primary package management and virtual environment tool, ensuring a smooth and well-integrated workflow for handling the numerous Python libraries required for this research. Furthermore, Microsoft Visual Studio Code (VS Code) has been utilized as the principal code editor, providing an efficient, customization, and user-friendly interface for writing, debugging, and executing the developed codes.

A significant part of the study relied on Qiskit, an open-source Software Development Kit (SDK) for quantum computing written in Python. Qiskit has been irreplaceable in constructing, simulating, and analyzing quantum circuits, as well as in executing them both on classical simulators and real quantum hardware. It has provided the necessary tools to translate theoretical quantum algorithms into executable codes.

Moreover, this work has benefited from the use of IBM's publicly available quantum computers, particularly *IBM Brisbane* and *IBM Torino*, which are accessible through the gratuitous cloud platform IBM Cloud. These devices enabled the experimental verification of theoretical results and offered an opportunity to study the effects of hardware noise and error mitigation strategies in real-world quantum processors.

I acknowledge the developers and maintainers of these software environments and platforms for making them freely available to the scientific community. Their contribution has been crucial in enabling research and experimentation in the field of quantum computing.

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Appendices

.1 Further calculations of analysis of Faulty Hadamard Gate in DJ Algorithm:

When 2nd hadamard of 1st layer is faulty:

The input state:

$$|\psi_0\rangle = |00\rangle |1\rangle$$

Operating $H_{faulty} \otimes H \otimes H$ on the input state:

$$|\psi_1\rangle = \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right)\left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right)\left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right)$$

$$\begin{aligned}|\psi_1\rangle = \frac{1}{2\sqrt{2}}\Big[&|00\rangle |0\rangle - |00\rangle |1\rangle + |01\rangle |0\rangle - |01\rangle |1\rangle \\ &- |10\rangle |0\rangle + |10\rangle |1\rangle - |11\rangle |0\rangle + |11\rangle |1\rangle\Big]\end{aligned}$$

Operating Oracle U_f on the state $|\psi_1\rangle$:

$$\begin{aligned}|\psi_2\rangle = \frac{1}{2\sqrt{2}}\Big[&|00\rangle |0 \oplus f(00)\rangle - |00\rangle |1 \oplus f(00)\rangle + |01\rangle |0 \oplus f(01)\rangle - |01\rangle |1 \oplus f(01)\rangle \\ &- |10\rangle |0 \oplus f(10)\rangle + |10\rangle |1 \oplus f(10)\rangle - |11\rangle |0 \oplus f(11)\rangle + |11\rangle |1 \oplus f(11)\rangle\Big]\end{aligned}$$

Bringing the global phase factor into account:

$$|\psi_2\rangle = \frac{1}{2}\left[(-1)^{f(00)} |00\rangle + (-1)^{f(01)} |01\rangle - (-1)^{f(10)} |10\rangle - (-1)^{f(11)} |11\rangle\right]\left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right)$$

Imposing the final layer of Hadamard gates:

$$\begin{aligned}|\psi_3\rangle = \frac{1}{2}\Bigg[&(-1)^{f(00)}\left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right)\left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right) + (-1)^{f(01)}\left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right)\left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right) \\ &-(-1)^{f(10)}\left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right)\left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right) - (-1)^{f(11)}\left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right)\left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right)\Bigg]\left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right)\end{aligned}$$

$$
\begin{aligned}
|\psi_3\rangle = \frac{1}{2}\Bigg[& |00\rangle \left((-1)^{f(00)} + (-1)^{f(01)} - (-1)^{f(10)} - (-1)^{f(11)}\right) \\
& + |01\rangle \left((-1)^{f(00)} - (-1)^{f(01)} - (-1)^{f(10)} + (-1)^{f(11)}\right) \\
& + |10\rangle \left((-1)^{f(00)} + (-1)^{f(01)} + (-1)^{f(10)} + (-1)^{f(11)}\right) \\
& + |11\rangle \left((-1)^{f(00)} - (-1)^{f(01)} + (-1)^{f(10)} - (-1)^{f(11)}\right) \Bigg] \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right)
\end{aligned}
$$

From the output state $|\psi_3\rangle$ we can deduce the following conclusions:
1. When function f is constant:

- f(00), f(01), f(10), f(11) all of these values can be either 0 or 1.
- In both of those cases, constructive quantum interference is seen for state $|10\rangle$ and destructive quantum interference is seen for rest of the states.

2. When the function f is balanced:

- Among output f(00), f(01), f(10) and f(11), any two values are 0 and the rest two values are 1.
- In that case constructive quantum interference can be seen for any state other than $|10\rangle$. Rest of the states suffers destructive interference.

When both hadamards of 1st layer are faulty:

The input state of Dj Algorithm for 2 qubit state:

$$|\psi_0\rangle = |00\rangle |1\rangle$$

Operating $H_{faulty} \otimes H \otimes H$ on the input state:

$$|\psi_1\rangle = \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right)$$

$$
\begin{aligned}
|\psi_1\rangle = \frac{1}{2\sqrt{2}}\Bigg[& |00\rangle |0\rangle - |00\rangle |1\rangle - |01\rangle |0\rangle + |01\rangle |1\rangle \\
& - |10\rangle |0\rangle + |10\rangle |1\rangle + |11\rangle |0\rangle - |11\rangle |1\rangle \Bigg]
\end{aligned}
$$

Operating Oracle U_f on the state $|\psi_1\rangle$:

$$
\begin{aligned}
|\psi_2\rangle = \frac{1}{2\sqrt{2}}\Bigg[& |00\rangle |0 \oplus f(00)\rangle - |00\rangle |1 \oplus f(00)\rangle - |01\rangle |0 \oplus f(01)\rangle + |01\rangle |1 \oplus f(01)\rangle \\
& - |10\rangle |0 \oplus f(10)\rangle + |10\rangle |1 \oplus f(10)\rangle + |11\rangle |0 \oplus f(11)\rangle - |11\rangle |1 \oplus f(11)\rangle \Bigg]
\end{aligned}
$$

Bringing the global phase factor into account:

$$|\psi_2\rangle = \frac{1}{2}\left[(-1)^{f(00)} |00\rangle - (-1)^{f(01)} |01\rangle - (-1)^{f(10)} |10\rangle + (-1)^{f(11)} |11\rangle\right] \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right)$$

Imposing the final layer of Hadamard gates:

$$
\begin{aligned}
|\psi_3\rangle = \frac{1}{2}\Bigg[&(-1)^{f(00)}\left(\frac{|0\rangle+|1\rangle}{\sqrt{2}}\right)\left(\frac{|0\rangle+|1\rangle}{\sqrt{2}}\right) - (-1)^{f(01)}\left(\frac{|0\rangle+|1\rangle}{\sqrt{2}}\right)\left(\frac{|0\rangle-|1\rangle}{\sqrt{2}}\right) \\
&-(-1)^{f(10)}\left(\frac{|0\rangle-|1\rangle}{\sqrt{2}}\right)\left(\frac{|0\rangle+|1\rangle}{\sqrt{2}}\right) + (-1)^{f(11)}\left(\frac{|0\rangle-|1\rangle}{\sqrt{2}}\right)\left(\frac{|0\rangle-|1\rangle}{\sqrt{2}}\right)\Bigg]\left(\frac{|0\rangle-|1\rangle}{\sqrt{2}}\right)
\end{aligned}
$$

$$
\begin{aligned}
|\psi_3\rangle = \frac{1}{2}\Bigg[&|00\rangle\left((-1)^{f(00)} - (-1)^{f(01)} - (-1)^{f(10)} + (-1)^{f(11)}\right) \\
&+|01\rangle\left((-1)^{f(00)} + (-1)^{f(01)} - (-1)^{f(10)} - (-1)^{f(11)}\right) \\
&+|10\rangle\left((-1)^{f(00)} - (-1)^{f(01)} + (-1)^{f(10)} - (-1)^{f(11)}\right) \\
&+|11\rangle\left((-1)^{f(00)} + (-1)^{f(01)} + (-1)^{f(10)} + (-1)^{f(11)}\right)\Bigg]\left(\frac{|0\rangle-|1\rangle}{\sqrt{2}}\right)
\end{aligned}
$$

From the output state $|\psi_3\rangle$ we can deduce the following conclusions:
1. When function f is constant:

- f(00), f(01), f(10), f(11) all of these values can be either 0 or 1.
- In both of those cases, constructive quantum interference is seen for state $|11\rangle$ and destructive quantum interference is seen for rest of the states.

2. When the function f is balanced:

- Among output f(00), f(01), f(10) and f(11), any two values are 0 and the rest two values are 1.
- In that case constructive quantum interference can be seen for any state other than $|11\rangle$. Rest of the states suffers destructive interference.

.2 Source Code:

Source code of this project can be found at the Github Repository: `https://github.com/rangan269-ju/Source-code-for-Summer-Project-on-Quantum-Algorithms.git`